

**DIFFUSION
MASS TRANSFER IN FLUID
SYSTEMS**

THIRD EDITION

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Models for Diffusion

If a few crystals of a colored material like copper sulfate are placed at the bottom of a tall bottle filled with water, the color will slowly spread through the bottle. At first the color will be concentrated in the bottom of the bottle. After a day it will penetrate upward a few centimeters. After several years the solution will appear homogeneous.

The process responsible for the movement of the colored material is diffusion, the subject of this book. Diffusion is caused by random molecular motion that leads to complete mixing. It can be a slow process. In gases, diffusion progresses at a rate of about 5 cm/min; in liquids, its rate is about 0.05 cm/min; in solids, its rate may be only about 0.00001 cm/min. In general, it varies less with temperature than do many other phenomena.

This slow rate of diffusion is responsible for its importance. In many cases, diffusion occurs sequentially with other phenomena. When it is the slowest step in the sequence, it limits the overall rate of the process. For example, diffusion often limits the efficiency of commercial distillations and the rate of industrial reactions using porous catalysts. It limits the speed with which acid and base react and the speed with which the human intestine absorbs nutrients. It controls the growth of microorganisms producing penicillin, the rate of the corrosion of steel, and the release of flavor from food.

In gases and liquids, the rates of these diffusion processes can often be accelerated by agitation. For example, the copper sulfate in the tall bottle can be completely mixed in a few minutes if the solution is stirred. This accelerated mixing is not due to diffusion alone, but to the combination of diffusion and stirring. Diffusion still depends on random molecular motions that take place over smaller distances. The agitation or stirring is not a molecular process, but a macroscopic process that moves portions of the fluid over much larger distances. After this macroscopic motion, diffusion mixes newly adjacent portions of the fluid. In other cases, such as the dispersal of pollutants, the agitation of wind or water produces effects qualitatively similar to diffusion; these effects, called dispersion, will be treated separately.

The description of diffusion involves a mathematical model based on a fundamental hypothesis or “law.” Interestingly, there are two common choices for such a law. The more fundamental, Fick’s law of diffusion, uses a diffusion coefficient. This is the law that is commonly cited in descriptions of diffusion. The second, which has no formal name, involves a mass transfer coefficient, a type of reversible rate constant.

Choosing between these two models is the subject of this chapter. Choosing Fick’s law leads to descriptions common to physics, physical chemistry, and biology. These descriptions are explored and extended in Chapters 2–7. Choosing mass transfer coefficients produces correlations developed explicitly in chemical engineering and used implicitly in chemical kinetics and in medicine. These correlations are described in Chapters 8–15. Both approaches are used in Chapters 16–21.

We discuss the differences between the two models in Section 1.1 of this chapter. In Section 1.2 we show how the choice of the most appropriate model is determined.

In Section 1.3 we conclude with additional examples to illustrate how the choice between the models is made.

1.1 The Two Basic Models

In this section we want to illustrate the two basic ways in which diffusion can be described. To do this, we first imagine two large bulbs connected by a long thin capillary (Fig. 1.1-1). The bulbs are at constant temperature and pressure and are of equal volumes. However, one bulb contains carbon dioxide, and the other is filled with nitrogen.

To find how fast these two gases will mix, we measure the concentration of carbon dioxide in the bulb that initially contains nitrogen. We make these measurements when only a trace of carbon dioxide has been transferred, and we find that the concentration of carbon dioxide varies linearly with time. From this, we know the amount transferred per unit time.

We want to analyze this amount transferred to determine physical properties that will be applicable not only to this experiment but also in other experiments. To do this, we first define the flux:

$$(\text{carbon dioxide flux}) = \left(\frac{\text{amount of gas removed}}{\text{time (area capillary)}} \right) \quad (1.1-1)$$

In other words, if we double the cross-sectional area, we expect the amount transported to double. Defining the flux in this way is a first step in removing the influences of our particular apparatus and making our results more general. We next assume that the flux is proportional to the gas concentration:

$$(\text{carbon dioxide flux}) = k \left(\frac{\text{carbon dioxide concentration difference}}{\text{distance}} \right) \quad (1.1-2)$$

The proportionality constant k is called a mass transfer coefficient. Its introduction signals one of the two basic models of diffusion. Alternatively, we can recognize

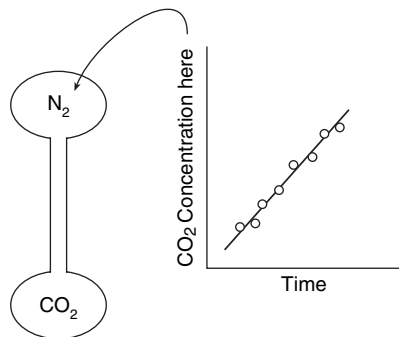


Fig. 1.1-1. A simple diffusion experiment. Two bulbs initially containing different gases are connected with a long thin capillary. The change of concentration in each bulb is a measure of diffusion and can be analyzed in two different ways.

that increasing the capillary's length will decrease the flux, and we can then assume that

$$(\text{carbon dioxide flux}) = D \left(\frac{\text{carbon dioxide concentration difference}}{\text{capillary length}} \right) \quad (1.1-3)$$

The new proportionality constant D is the diffusion coefficient. Its introduction implies the other model for diffusion, the model often called Fick's law.

These assumptions may seem arbitrary, but they are similar to those made in many other branches of science. For example, they are similar to those used in developing Ohm's law, which states that

$$\left(\begin{array}{c} \text{current, or} \\ \text{area times flux} \\ \text{of electrons} \end{array} \right) = \left(\frac{1}{\text{resistance}} \right) \left(\begin{array}{c} \text{voltage, or} \\ \text{potential} \\ \text{difference} \end{array} \right) \quad (1.1-4)$$

Thus, the mass transfer coefficient k is analogous to the reciprocal of the resistance. An alternative form of Ohm's law is

$$\left(\begin{array}{c} \text{current density} \\ \text{or flux of} \\ \text{electrons} \end{array} \right) = \left(\frac{1}{\text{resistivity}} \right) \left(\begin{array}{c} \text{potential} \\ \text{difference} \\ \text{length} \end{array} \right) \quad (1.1-5)$$

The diffusion coefficient D is analogous to the reciprocal of the resistivity.

Neither the equation using the mass transfer coefficient k nor that using the diffusion coefficient D is always successful. This is because of the assumptions made in their development. For example, the flux may not be proportional to the concentration difference if the capillary is very thin or if the two gases react. In the same way, Ohm's law is not always valid at very high voltages. But these cases are exceptions; both diffusion equations work well in most practical situations, just as Ohm's law does.

The parallels with Ohm's law also provide a clue about how the choice between diffusion models is made. The mass transfer coefficient in Eq. 1.1-2 and the resistance in Eq. 1.1-4 are simpler, best used for practical situations and rough measurements. The diffusion coefficient in Eq. 1.1-3 and the resistivity in Eq. 1.1-5 are more fundamental, involving physical properties like those found in handbooks. How these differences guide the choice between the two models is the subject of the next section.

1.2 Choosing Between the Two Models

The choice between the two models outlined in Section 1.1 represents a compromise between ambition and experimental resources. Obviously, we would like to express our results in the most general and fundamental ways possible. This suggests working with diffusion coefficients. However, in many cases, our experimental measurements will dictate a more approximate and phenomenological approach. Such approximations often imply mass transfer coefficients, but they usually still permit us to reach our research goals.

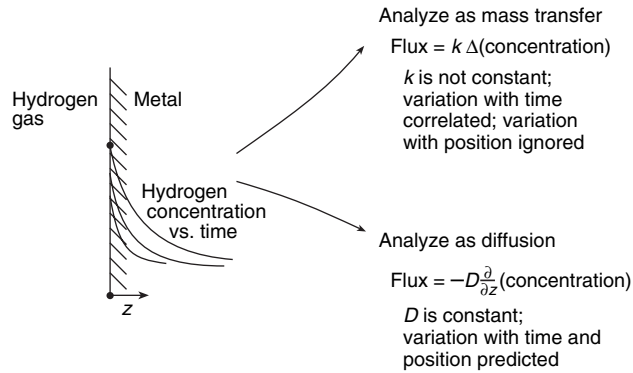


Fig. 1.2-1. Hydrogen diffusion into a metal. This process can be described with either a mass transfer coefficient k or a diffusion coefficient D . The description with a diffusion coefficient correctly predicts the variation of concentration with position and time, and so is superior.

This choice and the resulting approximations are best illustrated by two examples. In the first, we consider hydrogen diffusion in metals. This diffusion substantially reduces a metal's ductility, so much so that parts made from the embrittled metal frequently fracture. To study this embrittlement, we might expose the metal to hydrogen under a variety of conditions and measure the degree of embrittlement versus these conditions. Such empiricism would be a reasonable first approximation, but it would quickly flood us with uncorrelated information that would be difficult to use effectively.

As an improvement, we can undertake two sets of experiments. First, we can saturate metal samples with hydrogen and determine their degrees of embrittlement. Thus we know metal properties versus hydrogen concentration. Second, we can measure hydrogen uptake versus time, as suggested in Fig. 1.2-1, and correlate our measurements as mass transfer coefficients. Thus we know average hydrogen concentration versus time.

To our dismay, the mass transfer coefficients in this case will be difficult to interpret. They are anything but constant. At zero time, they approach infinity; at large time, they approach zero. At all times, they vary with the hydrogen concentration in the gas surrounding the metal. They are an inconvenient way to summarize our results. Moreover, the mass transfer coefficients give only the *average* hydrogen concentration in the metal. They ignore the fact that the hydrogen concentration very near the metal's surface will reach saturation but the concentration deep within the metal will remain zero. As a result, the metal near the surface may be very brittle but that within may be essentially unchanged.

We can include these details in the diffusion model described in the previous section. This model assumed that

$$\left(\begin{array}{c} \text{hydrogen} \\ \text{flux} \end{array} \right) = D \frac{\left(\begin{array}{c} \text{hydrogen} \\ \text{concentration at } z = 0 \end{array} \right) - \left(\begin{array}{c} \text{hydrogen} \\ \text{concentration at } z = l \end{array} \right)}{(\text{thickness at } z = l) - (\text{thickness at } z = 0)} \quad (1.2-1)$$

or, symbolically,

$$j_1 = D \frac{c_1|_{z=0} - c_1|_{z=l}}{l - 0} \quad (1.2-2)$$

where the subscript 1 symbolizes the diffusing species. In these equations, the distance l is that over which diffusion occurs. In the previous section, the length of the capillary was appropriately this distance; but in this case, it seems uncertain what the distance should be. If we assume that it is very small,

$$j_1 = D \lim_{l \rightarrow 0} \frac{c_1|_{z=z} - c_1|_{z=z+l}}{z|_{z+l} - z|_z} = -D \frac{dc_1}{dz} \quad (1.2-3)$$

We can use this relation and the techniques developed later in this book to correlate our experiments with only one parameter, the diffusion coefficient D . We then can correctly predict the hydrogen uptake versus time and the hydrogen concentration in the gas. As a dividend, we get the hydrogen concentration at all positions and times within the metal.

Thus the model based on the diffusion coefficient gives results of more fundamental value than the model based on mass transfer coefficients. In mathematical terms, the diffusion model is said to have distributed parameters, for the dependent variable (the concentration) is allowed to vary with all independent variables (like position and time). In contrast, the mass transfer model is said to have lumped parameters (like the average hydrogen concentration in the metal).

These results would appear to imply that the diffusion model is superior to the mass transfer model and so should always be used. However, in many interesting cases the models are equivalent. To illustrate this, imagine that we are studying the dissolution of a solid drug suspended in water, as schematically suggested by Fig. 1.2-2. The dissolution of this drug is known to be controlled by the diffusion of the dissolved drug away from the solid surface of the undissolved material. We measure the drug concentration versus time as shown, and we want to correlate these results in terms of as few parameters as possible.

One way to correlate the dissolution results is to use a mass transfer coefficient. To do this, we write a mass balance on the solution:

$$\begin{aligned} \left(\begin{array}{c} \text{accumulation} \\ \text{of drug in} \\ \text{solution} \end{array} \right) &= \left(\begin{array}{c} \text{total rate of} \\ \text{dissolution} \end{array} \right) \\ V \frac{dc_1}{dt} &= A j_1 \\ &= A k [c_1(\text{sat}) - c_1] \end{aligned} \quad (1.2-4)$$

where V is the volume of solution, A is the total area of the drug particles, $c_1(\text{sat})$ is the drug concentration at saturation and at the solid's surface, and c_1 is the concentration in the bulk solution. Integrating this equation allows quantitatively fitting our results with one parameter, the mass transfer coefficient k . This quantity is independent of drug solubility, drug area, and solution volume, but it does vary with physical properties like stirring rate and solution viscosity. Correlating the effects of these properties turns out to be straightforward.

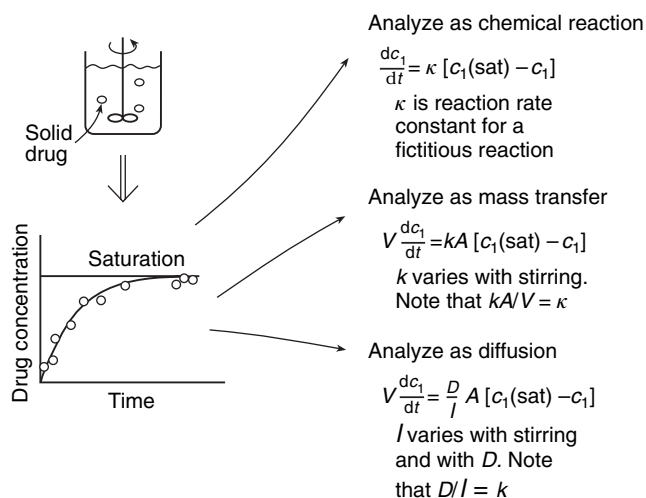


Fig. 1.2-2. Rates of drug dissolution. In this case, describing the system with a mass transfer coefficient k is best because it easily correlates the solution's concentration versus time. Describing the system with a diffusion coefficient D gives a similar correlation but introduces an unnecessary parameter, the film thickness l . Describing the system with a reaction rate constant k also works, but this rate constant is a function not of chemistry but of physics.

The alternative to mass transfer is diffusion theory, for which the mass balance is

$$V \frac{dc_1}{dt} = A \left(\frac{D}{l} \right) [c_1(\text{sat}) - c_1] \quad (1.2-5)$$

in which l is an unknown parameter, equal to the average distance across which diffusion occurs. This unknown, called a film or unstirred layer thickness, is a function not only of flow and viscosity but also of the diffusion coefficient itself.

Equations 1.2-4 and 1.2-5 are equivalent, and they share the same successes and shortcomings. In the former, we must determine the mass transfer coefficient experimentally; in the latter, we determine instead the thickness l . Those who like a scientific veneer prefer to measure l , for it genuflects toward Fick's law of diffusion. Those who are more pragmatic prefer explicitly recognizing the empirical nature of the mass transfer coefficient.

The choice between the mass transfer and diffusion models is thus often a question of taste rather than precision. The diffusion model is more fundamental and is appropriate when concentrations are measured or needed versus both position and time. The mass transfer model is simpler and more approximate and is especially useful when only average concentrations are involved. The additional examples in section 1.3 should help us decide which model is appropriate for our purposes.

Before going on to the next section, we should mention a third way to correlate the results other than the two diffusion models. This third way is to assume that the dissolution shown in Fig. 1.2-2 is a first-order, reversible chemical reaction. Such a reaction might be described by

$$\frac{dc_1}{dt} = \kappa c_1(\text{sat}) - \kappa c_1 \quad (1.2-6)$$

In this equation, the quantity $\kappa c_1(\text{sat})$ represents the rate of dissolution, κc_1 stands for the rate of precipitation, and κ is a rate constant for this process. This equation is mathematically identical with Eqs. 1.2-4 and 1.2-5 and so is equally successful. However, the idea of treating dissolution as a chemical reaction is flawed. Because the reaction is hypothetical, the rate constant is a composite of physical factors rather than chemical factors. We do better to consider the physical process in terms of a diffusion or mass transfer model.

1.3 Examples

In this section, we give examples that illustrate the choice between diffusion coefficients and mass transfer coefficients. This choice is often difficult, a juncture where many have trouble. I often do. I think my trouble comes from evolving research goals, from the fact that as I understand the problem better, the questions that I am trying to answer tend to change. I notice the same evolution in my peers, who routinely start work with one model and switch to the other model before the end of their research.

We shall not solve the following examples. Instead, we want only to discuss which diffusion model we would initially use for their solution. The examples given certainly do not cover all types of diffusion problems, but they are among those about which I have been asked in the last year.

Example 1.3-1: Ammonia scrubbing Ammonia, the major material for fertilizer, is made by reacting nitrogen and hydrogen under pressure. The product gas can be washed with water to dissolve the ammonia and separate it from other unreacted gases. How can you correlate the dissolution rate of ammonia during washing?

Solution The easiest way is to use mass transfer coefficients. If you use diffusion coefficients, you must somehow specify the distance across which diffusion occurs. This distance is unknown unless the detailed flows of gases and the water are known; they rarely are (see Chapters 8 and 9).

Example 1.3-2: Reactions in porous catalysts Many industrial reactions use catalysts containing small amounts of noble metals dispersed in a porous inert material like silica. The reactions on such a catalyst are sometimes slower in large pellets than in small ones. This is because the reagents take longer to diffuse into the pellet than they do to react. How should you model this effect?

Solution You should use diffusion coefficients to describe the simultaneous diffusion and reaction in the pores in the catalyst. You should not use mass transfer coefficients because you cannot easily include the effect of reaction (see Sections 16.1 and 17.1).

Example 1.3-3: Corrosion of marble Industrial pollutants in urban areas like Venice cause significant corrosion of marble statues. You want to study how these pollutants penetrate marble. Which diffusion model should you use?

Solution The model using diffusion coefficients is the only one that will allow you to predict pollutant concentration versus position in the marble. The model using

mass transfer coefficients will only correlate how much pollutant enters the statue, not what happens to the pollutant (see Sections 2.3 and 8.1).

Example 1.3-4: Protein size in solution You are studying a variety of proteins that you hope to purify and use as food supplements. You want to characterize the size of the proteins in solution. How can you use diffusion to do this?

Solution Your aim is determining the molecular size of the protein molecules. You are not interested in the protein mass transfer except as a route to these molecular properties. As a result, you should measure the protein's diffusion coefficient, not its mass transfer coefficient. The protein's diffusion coefficient will turn out to be proportional to its radius in solution (see Section 5.2).

Example 1.3-5: Antibiotic production Many drugs are made by fermentations in which microorganisms are grown in a huge stirred vat of a dilute nutrient solution or "beer." Many of these fermentations are aerobic, so the nutrient solution requires aeration. How should you model oxygen uptake in this type of solution?

Solution Practical models use mass transfer coefficients. The complexities of the problem, including changes in air bubble size, flow effects of the non-Newtonian solution, and foam caused by biological surfactants, all inhibit more careful study (see Chapter 8).

Example 1.3-6: Facilitated transport across membranes Some membranes contain a mobile carrier, a reactive species that reacts with diffusing solutes, facilitating their transport across the membrane. Such membranes can be used to concentrate copper ions from industrial waste and to remove carbon dioxide from coal gas. Diffusion across these membranes does not vary linearly with the concentration difference across them. The diffusion can be highly selective, but it is often easily poisoned. Should this diffusion be described with mass transfer coefficients or with diffusion coefficients?

Solution This system includes not only diffusion but also chemical reaction. Diffusion and reaction couple in a nonlinear way to give the unusual behavior observed. Understanding such behavior will certainly require the more fundamental model of diffusion coefficients (see Section 18.5).

Example 1.3-7: Flavor retention When food products are spray-dried, they lose a lot of flavor. However, they lose less than would be expected on the basis of the relative vapor pressures of water and the flavor compounds. The reason apparently is that the drying food often forms a tight gellike skin across which diffusion of the flavor compounds is inhibited. What diffusion model should you use to study this effect?

Solution Because spray-drying is a complex, industrial-scale process, it is usually modeled using mass transfer coefficients. However, in this case you are interested in the inhibition of diffusion. Such inhibition will involve the sizes of pores in the food and of molecules of the flavor compounds. Thus you should use the more basic diffusion model, which includes these molecular factors (see Section 6.4).

Example 1.3-8: The smell of marijuana Recently, a large shipment of marijuana was seized in the Minneapolis–St. Paul airport. The police said their dog smelled it. The owners claimed that it was too well wrapped in plastic to smell and that the police had conducted an illegal search without a search warrant. How could you tell who was right?

Solution In this case, you are concerned with the diffusion of odor across the thin plastic film. The diffusion rate is well described by either mass transfer or diffusion coefficients. However, the diffusion model explicitly isolates the effect of the solubility of the smell in the film, which dominates the transport. This solubility is the dominant variable (see Section 2.2). In this case, the search was illegal.

Example 1.3-9: Scale-up of wet scrubbers You want to use a wet scrubber to remove sulfur oxides from the flue gas of a large power plant. A wet scrubber is essentially a large piece of pipe set on its end and filled with inert ceramic material. You pump the flue gas up from the bottom of the pipe and pour a lime slurry down from the top. In the scrubber, there are various reactions, such as



The lime reacts with the sulfur oxides to make an insoluble precipitate, which is discarded. You have been studying a small unit and want to use these results to predict the behavior of a larger unit. Such an increase in size is called a scale-up. Should you make these predictions using a model based on diffusion or mass transfer coefficients?

Solution This situation is complex because of the chemical reactions and the irregular flows within the scrubber. Your first try at correlating your data should be a model based on mass transfer coefficients. Should these correlations prove unreliable, you may be forced to use the more difficult diffusion model (see Chapters 9, 16, and 17).

1.4 Conclusions

This chapter discusses the two common models used to describe diffusion and suggests how you can choose between these models. For fundamental studies where you want to know concentration versus position and time, use diffusion coefficients. For practical problems where you want to use one experiment to tell how a similar one will behave, use mass transfer coefficients. The former approach is the distributed-parameter model used in chemistry, and the latter is the lumped-parameter model used in engineering. Both approaches are used in medicine and biology, but not always explicitly.

The rest of this book is organized in terms of these two models. Chapters 2–4 present the basic model of diffusion coefficients, and Chapters 5–7 review the values of the diffusion coefficients themselves. Chapters 8–15 discuss the model of mass transfer coefficients, including their relation to diffusion coefficients. Chapters 16–19 explore the coupling of diffusion with heterogeneous and homogeneous chemical reactions, using both models. Chapters 20–21 explore the simpler coupling between diffusion and heat transfer.

In the following chapters, keep both models in mind. People involved in basic research tend to be overcommitted to diffusion coefficients, whereas those with broader objectives tend to emphasize mass transfer coefficients. Each group should recognize that the other has a complementary approach that may be more helpful for the case in hand.

Questions for Discussion

1. What are the dimensions in mass M , length L , and time t of a diffusion coefficient?
2. What are the dimensions of a mass transfer coefficient?
3. What volume is implied by Fick's law?
4. What volume is implied when defining a mass transfer coefficient?
5. Can the diffusion coefficient ever be negative?
6. Give an example for a diffusion coefficient which is the same in all directions. Give an example when it isn't.
7. When a silicon chip is doped with boron, does the doping involve diffusion?
8. Does the wafting of smells of a pie baking in the oven involve diffusion?
9. How does breathing involve diffusion?
10. How is a mass transfer coefficient related to a reaction rate constant?
11. Will a heat transfer coefficient and a mass transfer coefficient be related?
12. Will stirring a suspension of sugar in water change the diffusion coefficient? Will it change the density? Will it change the mass transfer coefficient?